0. **Warm-up:** When you solve the linear equation, you must be able to answer all the sub-question below.

Given $\mathbf{A} = \begin{bmatrix} 2 & 2 & -2 & 1 & -2 \\ 3 & 1 & 0 & 0 & 1 \\ 1 & -1 & 2 & 0 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

a. If $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{x} \in \mathbf{R}^n$, then $n =$ _____

b. If $\mathbf{A}$ represents linear map T: $\mathbf{R}^m \Rightarrow \mathbf{R}^n$, then $n =$ _____, and $m =$ _____

c. This equation is homogeneous or inhomogeneous, justify your answer.

d. Rewrite $\mathbf{Ax} = \mathbf{b}$ as a linear combination of the columns of $\mathbf{A}$.

e. Rewrite $\mathbf{Ax} = \mathbf{b}$ as a system of equations (using dot product).

f. Find the coefficient matrix and augmented matrix.

g. From f., find the RRE form of the coefficient matrix.

h. Pivot variable(s): _____

i. Free variable(s): _____

j. How many solutions are there?

k. If no solutions, why? If there is/are solution(s), then find the **solution set.**

l. Find the particular part and homogeneous part.

m. Is T one-to-one? Is T onto? Briefly explain.

1. Given a matrix $\mathbf{A} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$. Find one eigenvector of $\mathbf{A}$ and its eigenvalue.

   [This is one of the tricky problems from many years ago **mid-term** exam.]

2. By performing a row operation on a matrix, one can say that a matrix goes through a matrix multiplication. As a result, every step of row operation can always be described with a matrix multiplication.

   For example, if we started with a matrix $\mathbf{B} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and we would like to perform a row-swap on row 1 and 2 (Equivalently, $\mathbf{B} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{row-swap}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

   then we can say that $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a matrix represents this row-swap operations as

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   Now, try finding the matrix that represents the following row-operation.

   2.1. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \leftarrow 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 2\cdot 4 & 2\cdot 5 & 2\cdot 6 \\ 7 & 8 & 9 \end{bmatrix}$

   2.2. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_3 \leftarrow 3R_1 + R_3} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3\cdot 1 + 7 & 3\cdot 2 + 8 & 3\cdot 3 + 9 \end{bmatrix}$

2.3. Let $\mathbf{A} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$ and $\mathbf{A}_r$ represents the reduced row echelon form of $\mathbf{A}$.

Find matrix $\mathbf{C}$ such that $\mathbf{CA} = \mathbf{A}_r$.

3. Given two linear transformations T: $\mathbf{R}^3 \Rightarrow \mathbf{R}^2$ and S: $\mathbf{R}^2 \Rightarrow \mathbf{R}^3$ which represented by matrices $\mathbf{A}$ and $\mathbf{B}$ respectively. Answer the following questions.

3.1. Can $T \circ S$ be one-to-one or onto? justify your answer.

3.2. Can $S \circ T$ be one-to-one or onto? justify your answer.

3.3. If both RRE($\mathbf{A}$) and RRE($\mathbf{B}$) have two pivots, is $T \circ S$ always isomorphic?

4. Let L be a line through $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, in the $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$ direction.

4.1. Find the augmented matrix in RRE form which has L **as its solution**.

4.2. Find the smallest distance from $\begin{bmatrix}0\\0\\0\end{bmatrix}$ to L.

5. Find a **linear equation system** describing the following sub-questions.

5.1. Line containing $\begin{bmatrix}1\\2\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\\1\end{bmatrix}$

5.2. Plane containing $\begin{bmatrix}-2\\0\\1\\-1\end{bmatrix}$, $\begin{bmatrix}4\\0\\0\\4\end{bmatrix}$ and $\begin{bmatrix}1\\2\\1\\0\end{bmatrix}$

6. Let $P = \left\{ \begin{bmatrix}0\\0\\1\\-1\end{bmatrix} + \begin{bmatrix}\text{-1}\\2\\0\\1\end{bmatrix} a + \begin{bmatrix}1\\2\\0\\1\end{bmatrix} b \;\middle|\; a, b \in \mathbf{R} \right\}$ and $Q = \left\{ \begin{bmatrix}x\\y\\z\\w\end{bmatrix} \in \mathbf{R}^4 \;\middle|\; w - y - z + x = 0 \right\}$

6.1. Find S where $S = P \cap Q$

6.2. Let T be parallel to S but pass through the origin, find T.

7. Find the determinant and the inverses of the following matrices.

7.1. $$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 3 \\ 0 & 0 & 2 & -2 & 3 \\ 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & 3 & 1 & 4 \end{bmatrix}^T$$

7.2. $$\left(\left(\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}^3 \begin{bmatrix} 0 & 1 \\ 1 & \pi \end{bmatrix}\right)^{-1} 3 \begin{bmatrix} 1 & 0 \\ \sqrt{5} & 2 \end{bmatrix}\right)^T$$

8. Find a span description for the intersection of the hyperplanes $x_1 + x_3 + x_5 = 0$ and $x_2 + x_4 = 0$ in $\mathbf{R}^5$.

9. Show that a set of k vectors in $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbf{R}^n$ **cannot be independent** if $k > n$.

10. Given $\mathbf{A} = \begin{bmatrix} 2 & 4 & 0 & 6 \\ 4 & 8 & 2 & 6 \end{bmatrix}$. Find a matrix $\mathbf{B}$ whose row-space is the left-nullspace of $\mathbf{A}$.